Mobile Networks and Applications (2024) 29:1378–1392
https://doi.org/10.1007/s11036-024-02411-y

RESEARCH

TMPSformer: An Efficient Hybrid Transformer-MLP Network for Polyp Segmentation

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Accepted: 20 August 2024 / Published online: 10 September 2024

Abstract
Colorectal cancer poses a global health risk, often heralded by colorectal polyps. Colonoscopy is the primary modality for polyp detection, with precise, real-time segmentation being key to effective diagnosis and surgical planning. Existing segmentation models like convolutional neural networks (CNNs) and Transformers have propelled progress but face trade-offs between precision and speed. CNNs excel in local feature extraction yet struggle with global context, while Transformers handle global information well but at a computational cost. Addressing these constraints, we introduce TMPSformer, a groundbreaking lightweight model tailored for efficient and accurate real-time polyp segmentation. TMPSformer, with its compact size of only 2.7 M, features a pioneering hybrid encoder merging Transformers' long-range dependencies and shift Multi-Layer Perceptrons (MLPs)' local dependencies, effectively enhancing segmentation performance. It also equips an All-MLP decoder to streamline feature fusion and enhance decoding efficiency. TMPSformer utilizes the Flash Efficient Attention (FEA) module to replace the traditional Attention module, significantly improving real-time performance. A comprehensive evaluation on five public polyp segmentation datasets demonstrated TMPSformer's superiority over existing state-of-the-art algorithms. Specifically, TMPSformer achieves real-time processing at 162 frames per second (FPS) at 512×512 resolution on the Kvasir-SEG dataset using a single NVIDIA RTX 2080 Ti GPU, and achieves a mean Intersection over Union (mIoU) of 0.811. Its segmentation performance surpasses ColonSegNet by 8.7% and SegFormer by 4.8%. Additionally, TMPSformer significantly reduces complexity, cutting the parameter count by 1.8× and 31× compared to ColonSegNet and SegFormer, respectively.

Keywords Tansformer · MLP · Polyp segmentation · Cancer diagnosis · Deep learning

1 Introduction

Colorectal Cancer (CRC) is the third most prevalent form of cancer worldwide, with an estimated 1.9 million new cases and over 930,000 deaths in 2020 alone. By 2040, the burden of CRC is projected to increase to 3.2 million new cases and 1.6 million deaths annually, highlighting the urgent need for effective early diagnosis and treatment strategies [1]. Early diagnosis is a critical factor in improving patient survival rates, as it significantly enhances the chances of successful treatment and long-term survival [2]. Colonoscopy is the standard diagnostic tool for CRC screening, utilizing an endoscope to inspect the intestines for polyps, which can potentially develop into CRC. However, a significant challenge in colonoscopy is the high rate of missed polyp diagnoses, primarily due to the variability in physician expertise [3]. To address this issue, there has been a growing interest in automated polyp segmentation approaches. These aim to autonomously identify polyps in colonoscopy images, reducing reliance on physician interpretation. However, the inherent challenges, such as the low contrast and diverse shapes of polyps, make automated segmentation a daunting task.

Deep learning has revolutionized numerous fields due to its ability to automatically learn and extract features from data, leading to its widespread application in various

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domains [4–8]. Given the power of deep learning technologies, an increasing number of researchers have applied these techniques to medical image segmentation tasks, achieving significant progress. Particularly, Convolutional Neural Network (CNN)-based methods like UNet [9], with its encoder-decoder framework and skip connections, have been widely adopted. Its success has inspired several variants, including UNet++ [10], ResUNet [11], Y-Net [12], V-Net [13], and UNet3+ [14]. Despite their effectiveness in medical image segmentation, these models often struggle with overfitting, especially on smaller datasets, and the reduction of spatial resolution during encoding can result in the loss of critical local features, impacting segmentation accuracy.

In parallel, the Transformer [15], originally developed for natural language processing, has shown great potential in computer vision. Its ability for global information representation has led to its adoption in medical image segmentation, as seen in TransUNet [16], UNETR [17], and CoTr [18], which combine Transformer and CNN architectures. However, the computational intensity of the Transformer's self-attention mechanism remains a challenge for real-time applications. This limitation has led to an increased interest in Multilayer Perceptron (MLP) based methods, like AS-MLP [19], which offer computational efficiency and competitive performance but are less explored in medical image segmentation. Addressing these challenges, we introduce an innovative network for efficient and accurate real-time polyp segmentation, named **TMPSformer**, which leverages the complementary strengths of Transformer and shift MLP architectures. This model aims to provide a high-accuracy segmentation tool that functions efficiently in real-time polyp segmentation environments without requiring extensive computational resources or specialized hardware. Our newly developed method surpasses other state-of-the-art (SOTA) approaches, delivering superior segmentation performance and extremely low computational complexity. This enhances its practicality for efficient polyp segmentation and supports early CRC detection and treatment.

The contributions of our paper are threefold, summarized as follows. (1) We present TMPSformer, an innovative hybrid lightweight segmentation network architecture tailored for the real-time and precise segmentation of polyps. This network synergistically integrates Transformer and shift MLP architectures, harnessing their respective strengths to effectively capture global and local semantic features for accurate segmentation mask prediction. It also innovatively utilizes deep separable convolution operations as an alternative to the Transformer's position encoding, resulting in a more streamlined and faster network. (2) Our method introduces the Flash Efficient Attention (FEA) module, which leverages spatial reduction attention (SRA) and a memory read/write minimization technique from FlashAttention to replace the traditional Attention module in the Transformer architecture, significantly enhancing the efficiency of the segmentation process. (3) We evaluate our model on challenging polyp segmentation datasets, benchmarking it against SOTA methods. Our findings underscore the model's real-time capabilities and superior segmentation performance, despite having fewer parameters, i.e., with only 2.7 M.

2 Related Works

2.1 Traditional Image Feature-Based Methods

Initial research in polyp segmentation primarily focused on extracting hand-crafted image features, such as color, shape, and texture, to identify and segment polyps. Yu et al. [20] represents a notable example of this approach. However, these methods often struggled with the heterogeneous nature of polyps and their resemblance to surrounding tissues, leading to limited accuracy. With the advent of deep learning, data-driven AI methods have significantly improved polyp segmentation. These can be broadly categorized into CNN-based, MLP-based, and Transformer-based approaches.

2.2 CNN-Based Approaches

The introduction of Fully Convolutional Networks (FCNs) by Long et al. [21] marked a pivotal shift toward end-to-end trainable segmentation models. This was further advanced by Ronneberger et al. [9] with the UNet architecture, which integrates deep and shallow image features through a symmetrical U-shaped encoder-decoder structure and skip connections. Building on UNet's success, various enhancements like DeepLabV3+ [22] have been introduced, employing atrous separable convolutions and spatial pyramids for faster inference and improved accuracy. Subsequent innovations include ResUNet++ [23] and DoubleU-Net [24], which incorporate additional layers like squeeze-and-excite blocks [25] and Atrous Spatial PyramidPooling (ASPP) [26]. These modifications have been instrumental in enhancing the network's ability to discern objects at different scales. Additionally, models like PraNet [27], PolypSeg [28], and others have integrated specialized modules for deeper feature learning, thereby improving segmentation accuracy.

In the realm of video polyp segmentation, CNN-based methods also have shown promising results. A notable example is the hybrid 2/3D CNN framework [3], which effectively captures spatio-temporal correlations, demonstrating substantial performance in video polyp segmentation tasks.

2.3 MLP-Based Approaches

Recently, MLP-based networks [19, 29–31] have demonstrated their efficacy in computer vision tasks. AS-MLP [19], an all-MLP network, stands out for its reduced computational needs while maintaining competitive performance. Inspired by this, UNeXt [32] was introduced for point-of-care imaging, blending convolution and MLP blocks in a UNet-like architecture. This approach not only reduces model parameters and inference time but also enhances segmentation performance.

2.4 Transformer-Based Approaches

The advent of Transformer architectures offers powerful global modeling capabilities. This has led researchers to increasingly adopt Transformer-based methods, resulting in significant advancements in various medical image analysis tasks. One notable example is TransFuse [33], which employs a parallel branch scheme combining Transformer and CNN to simultaneously capture global dependencies and low-level spatial details. Similarly, Segtran [34] introduces a compressed attention block that normalizes self-attention, enabling the learning of diverse representations. The groundbreaking work of ViT [35] demonstrated that a pure Transformer could achieve state-of-the-art performance in image classification, a milestone in the evolution of these models.

In the domain of 2D medical image segmentation, TransUNet [16] adapted the ViT architecture into a UNet framework, illustrating the versatility of Transformers in this field. Further extending the application of Transformer architectures, networks like MedT [36], TransBTS [37], and UNETR [17] have been developed, each contributing to the evolving landscape of medical image segmentation. Additionally, self-attention-based methods have found applications in video polyp segmentation. PNSNet [38], as a prime example, introduced normalized self-attention blocks that effectively learn spatio-temporal representations, demonstrating enhanced segmentation performance.

While Transformer-based methods offer unparalleled advantages in modeling global relationships, they often suffer from low computational efficiency and limited local feature extraction capabilities. These drawbacks have restricted their applicability in real-time polyp segmentation scenarios, which emphasize both global and local feature dependencies. To address these challenges, our paper introduces an innovative network that leverages the strengths of Transformer and shift MLP architectures, specifically tailored for efficient and effective polyp segmentation tasks.

3 Method

This section elaborates on the architecture and functioning of TMPSformer. TMPSformer distinguishes itself through its unique combination of encoder and decoder, strategically optimized for high efficiency and performance.

3.1 Overview of TMPSformer

At its core, TMPSformer features an innovative structure composed of an encoder and a lightweight MLP decoder. The encoder is a fusion of a hierarchical Transformer stage and a shift MLP stage. This design facilitates the generation of multi-level semantic features: coarse features at high resolution and fine features at low resolution. This design allows the model to capture and integrate features at different scales effectively. Specifically, the hierarchical structure enables the extraction of coarse, high-level semantic information from higher resolution layers, which is crucial for understanding the overall context and structure of the image. Simultaneously, it extracts fine, detailed features from lower resolution layers, which are essential for precise boundary delineation and local detail representation. The decoder then integrates these multi-scale features, combining the global context with local details to produce a comprehensive and accurate semantic segmentation mask, as illustrated in Fig. 1. By leveraging this hierarchical approach, TMPSformer ensures that both global and local features are effectively extracted and fused, enhancing the overall segmentation performance.

3.2 Encoder Design

The encoder of TMPSformer integrates Transformer and shift MLP components. This arrangement efficiently generates semantic features at multiple levels, enhancing the network's segmentation capability.

Given an input image of dimensions $H \times W \times 3$, the image is first partitioned into patches of size 4×4. These patches sequentially pass through two Transformer blocks and two shift MLP blocks. This process yields hierarchical features at resolutions of $\{\frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}\}$ of the original image size, which are subsequently fused. The output channels for these blocks are denotes as C_1, C_2, C_3, and C_4, with the experimental setup utilizing $C_1 = 32$, $C_2 = 64$, $C_3 = 160$, and $C_4 = 256$.

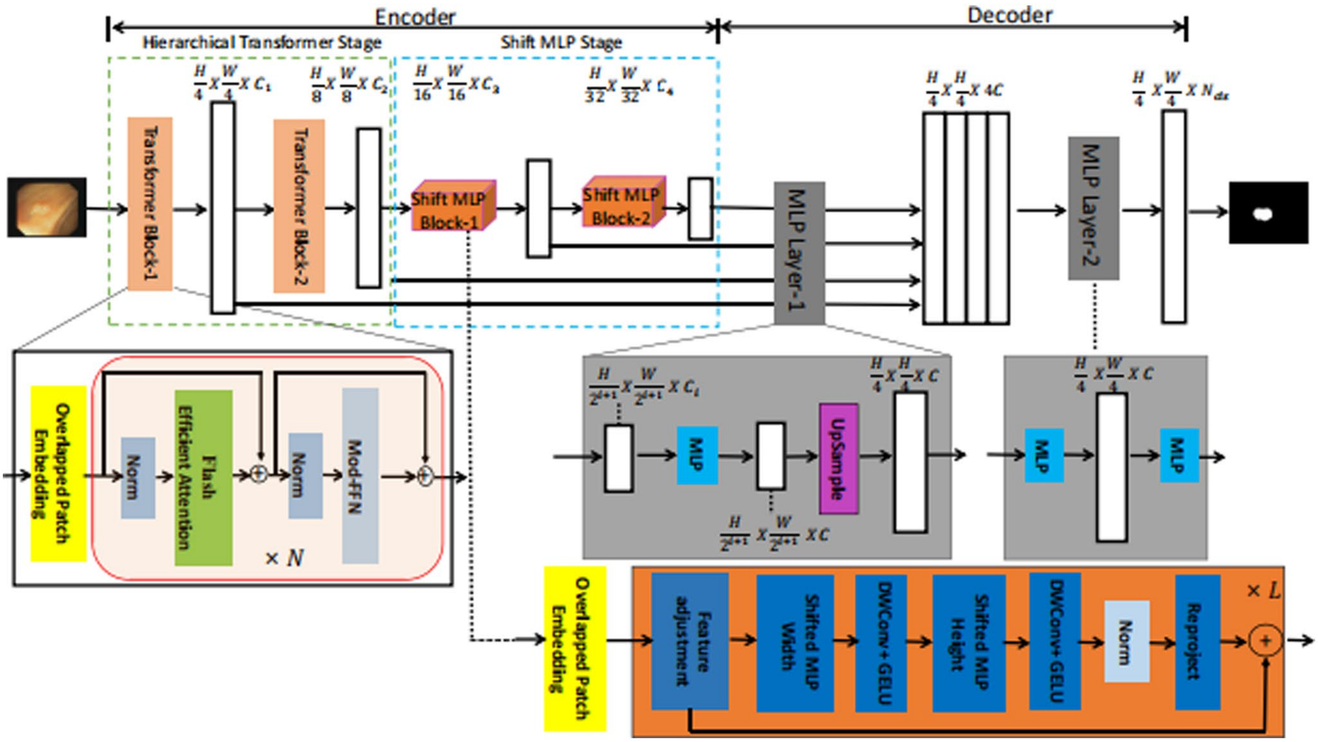

Fig. 1 Overall architecture of TMPSformer. The encoder uses a hierarchical transformer stage to process input features. The shift MLP stage then applies two sequential shift MLP blocks to further transform the features. Finally, the decoder reconstructs the output using MLP layers, and produces the final result

3.2.1 Hierarchical Transformer

The hierarchical Transformer stage is composed of two transformer blocks, each consisting of Overlapped Patch Embedding, Flash Efficient Attention (**FEA**), and Modified Feed-Forward Network (**Mod-FFN**). In each transformer block, Overlapped Patch Embedding is used once, followed by FEA and Mod-FFN, which are repeated N times with skip connections. In the experiment, the two transformer blocks are configured with N, which is a hyperparameter, set as follows: Transformer block-1 with ($N_1=3$), and Transformer block-2 with ($N_2=6$).

Overlapped Patch Embedding Our approach innovates upon the standard ViT by merging non-overlapping feature patches to maintain local feature coherence. Unlike ViT, which can only generate a single-resolution feature map, our encoder aims to generate CNN-like multi-level features from an input image. In ViT, during patch embedding, the image is divided into non-overlapping patches for encoding, which fails to preserve the local continuity around those patches. To address this, we draw inspiration from convolutional operations and divide the image into overlapping patches during patch embedding to enhance local feature extraction. This embedding of overlapped patches can be easily implemented using convolution operations. The degree of overlap is controlled by the parameters K_e (patch size), S_t (stride), and P_a (padding size). In our experiment, K_e, S_t, and P_a are set to 7, 4, and 3 in the first transformer block, and 3, 2, and 1 in the second transformer block, respectively. Through the overlapped patch embedding operation in the transformer blocks, we obtain hierarchical encoded features: Transformer block-1 produces F_1 ($\frac{H}{4} \times \frac{W}{4} \times C_1$), and Transformer block-2 produces $F_2(\frac{H}{8} \times \frac{W}{8} \times C_2)$. In the subsequent shift MLP blocks, we further utilize overlapped patch embedding to obtain hierarchical features from $F_3(\frac{H}{16} \times \frac{W}{16} \times C_3)$ to $F_4(\frac{H}{32} \times \frac{W}{32} \times C_4$). This effectively enhances the encoder's ability to extract local information.

Flash Efficient Attention Addressing the computational bottleneck of the attention layer in transformers, we combine middle dimensionality reduction and memory optimization techniques to enhance computational efficiency. Concretely, the conventional self-attention mechanism is quantified by Eq. (1).

$$Attention(Q,K,V) = Softmax\left(\frac{Q \cdot K^T}{\sqrt{d}}\right) \cdot V \tag{1}$$

where Q, K, and V denote the query, key, and value components of the attention mechanism, and d is the dimension of the keys and queries. Given a sequence length N, the computational complexity of self-attention is $O(N^2)$, which can be prohibitively expensive for high-resolution colonoscopy images. To overcome this, our TMPSformer model employs FEA modules, which combine two advanced techniques: a middle dimensionality reduction method—SRA [39] and a memory read/write minimization technique from FlashAttention [40]. The FEA module is designed to enhance the efficiency and performance of the segmentation process.

SRA addresses the high computational and memory costs associated with traditional self-attention mechanisms by applying 2D convolution layers to downsample the hidden states (queries, keys, and values) before the attention operation. Additionally, the FlashAttention technique minimizes memory reads and writes by dividing the input sequences into smaller blocks or tiles and processing them sequentially. This IO-aware approach optimizes data movement between different levels of GPU memory, reducing IO complexity and ensuring efficient utilization of the GPU's memory hierarchy. By avoiding the storage of large intermediate attention matrices in high-bandwidth memory (HBM), FlashAttention lowers memory usage and speeds up computation.

The integration of these two techniques in the FEA module decreases the computational complexity from quadratic to linear in sequence length, making the model more efficient for dense prediction tasks such as segmentation. It significantly reduces computational and memory costs, leading to faster training and inference times. The memory savings allow the TMPSformer model to handle longer sequences and larger batch sizes without running into memory limitations. The modified attention calculation is articulated through Eq. (2) to (4).

$$K' = Reshape\left(\frac{N}{R_a}, C \cdot R_a\right)(K) \tag{2}$$

$$K = Linear(C \cdot R_a, C)(K') \tag{3}$$

$$SA = FalshAttention(Q, K, V) \tag{4}$$

Here, R_a denotes the dimensionality reduction ratio, which reconfigures the key sequence K into a lower-dimensional representation, effectively reducing the computational complexity of self-attention from $O(N^2)$ to $O(\frac{N^2}{R_a})$. The sequence is then linearly transformed from the reduced dimensionality back to the original space, preserving essential information while minimizing computational expense. The Flash Attention operation is subsequently applied to compute the self-attention value SA, significantly accelerating the process. In our experimental setup, we have assigned $R_a = 2$ for the first transformer block and $R_a = 4$ for the second, facilitating enhanced efficiency for real-time applications.

Mod-FFN In the TMPSformer framework, the Mod-FFN plays a critical role in refining the feature processing pipeline. Drawing inspiration from the SegFormer [41], the Mod-FFN replaces positional encoding with a 3×3 convolutional operation within the FFN. This approach is encapsulated by the expression outlined in Eq. (5).

$$X_{out} = MLP\left(GELU\left(Conv\,3\times3\left(MLP\left(LN\left(X_{in}\right)\right)\right)\right)\right) + X_{in} \tag{5}$$

where LN represents Layer Normalization, and X_{in} is the resultant feature map post self-attention processing. The Mod-FFN ingeniously fuses a 3×3 convolution with an MLP layer in each FFN, utilizing the Gaussian Error Linear Unit (GELU) [42] as the activation function. For our empirical analysis, a 3×3 depthwise separable convolution [43] is employed instead of the traditional 3×3 convolution to streamline the network by reducing parameter count. Depthwise separable convolution has become a popular alternative to traditional convolutions due to its ability to significantly reduce the number of parameters and computational costs, without degrading performance. Traditional convolutional layers perform both spatial and channel-wise computations in a single step, which can be computationally expensive. In contrast, depthwise separable convolutions split this process into two steps: depthwise convolution and pointwise convolution. The depthwise convolution applies a single filter per input channel, while the pointwise convolution uses a 1×1 filter to combine the outputs of the depthwise convolution. This separation reduces the number of parameters and floating-point operations (FLOPs) required. By utilizing depthwise separable convolution, our model enhances computational efficiency without compromising performance.

3.2.2 Shift MLP

Within the TMPSformer architecture, the shift MLP stage is an innovative component designed to exact fine features at lower resolutions. This stage consists of two shift MLP blocks, each tasked with producing low-resolution, yet detailed features through a forth-part process: Overlapped Patch Embedding, Feature Adjustment, Shifted MLP Height, and Shifted MLP Width. In each shift MLP block, an overlapped patch embedding, similar to that in the transformer blocks, is applied once. This is followed by feature adjustment, and then shifted MLP height and shifted MLP width, which are repeated L times. In the experiment, the two shift MLP blocks are configured with L set as follows:

shift MLP block-1 with ($L_1 = 2$), and shift MLP block-2 with ($L_2 = 2$).

Feature Adjustment The Feature Adjustment process commences with the application of layer normalization to the outputs of the Transformer, denoted by *T*. This step normalizes the output across the each feature for a given layer, stabilizing the learning process. Subsequently, the normalized feature maps are reshaped to prepare them for further processing. The operation is mathematically represented by Eq. (6).

$$Fr = Reshape\left(LN\left(T\right)\right) \tag{6}$$

where *Fr* signifies the reshaped feature tensor, and *T* is the original Transformer output with dimensions (B, $H \times W$, C_T), and B is the batch size, $H \times W$ is the spatial dimensions, and C_T is the number of channels. This restructured feature set *Fr*, with dimensions (B, C_T, H, W), is then poised for subsequent stages of feature transformation within the shift MLP blocks.

Shifted MLP Height and Width The process begins with the shifted MLP height, where we adjust the spatial arrangement of the feature map's height dimension. This initial step is motivated by the concept of axial-attention [44] and the Swin Transformer's window-based attention mechanism [45], which introduces more locality to the traditionally global perspective of the model. Specifically, we apply a shift operation along the height axis on the normalized feature maps *(Fr)*, segmenting the features into *(p)* partitions and shifting each by *(s)* positions. This operation ensures that the subsequent MLP layer focuses on specific feature locations.

According to reference [19], we set $p = 3$ and $s = 2$ in the experiments. Following the spatial adjustment, the features undergo tokenization and dimension conversion through an MLP, from C_T to $\widehat{C_T}$. The transformed feature tokens are then processed by a 3×3 convolutional operation, utilizing the GELU activation function. This convolutional approach serves as an alternative to traditional positional encoding, similar to the operation in the transformer block, focusing on capturing spatial relationships within the features. The shifted MLP width mirrors the procedure of the shifted MLP height, albeit with a focus on the width axis. This symmetrical treatment of height and width dimensions ensures a comprehensive coverage of spatial variations within the image. The full procedure is detailed from Eq. (7) to (13).

$$F_{shift} = Shift_H\left(Fr\right) \tag{7}$$

$$T_H = Tokenize\left(F_{shift}\right) \tag{8}$$

$$Y_H = GELU\left(Conv\,3 \times 3\left(MLP\left(T_H\right)\right)\right) \tag{9}$$

$$Y_{shift} = Shift_W\left(Y_H\right) \tag{10}$$

$$T_W = Tokenize\left(Y_{shift}\right) \tag{11}$$

$$Y = GELU\left(Conv\,3 \times 3\left(MLP\left(T_W\right)\right)\right) \tag{12}$$

$$Y_{out} = Y + LN\left(T\right) \tag{13}$$

where *Fr* denotes the reshaped features, *T* the original transformer outputs, and $Conv\ 3 \times 3$ represents a depthwise separable convolution utilized for spatial feature processing.

Through the shifted MLP height and width operations, the feature maps are segmented into (p) partitions and shifted along both the height and width dimensions to form a cross-shaped local receptive field [19]. This configuration is highly beneficial for extracting more fine local detail features.

3.3 Decoder

The decoder of the TMPSformer utilizes an all-MLP design, setting a benchmark for computational efficiency compared to conventional methods. This streamlined decoder comprises two primary MLP layers, each playing a crucial role in processing and refining the features extracted by the encoder for accurate polyp segmentation.

MLP Layer-1 The first MLP layer serves as the initial processing stage for the multi-level features F_i from the encoder. These features undergo a channel unification step through a linear transformation, aligning them for subsequent operations. Following this, the features are upsampled to a quarter of the original resolution and concatenated. The operations conducted within this layer are defined in Equations (14) to (15).

$$F_i' = Linear\left(C_i, C\right)\left(F_i\right), \forall\, i \tag{14}$$

$$F'' = Concat\left(Upsample\left(\frac{H}{4} \times \frac{W}{4}\right)\left(F_i'\right)\right), \forall\, i \tag{15}$$

MLP Layer-2 The concatenated feature set, after being processed through an additional MLP layer dedicated to fusion, results in F, which enhances the coherence and spatial alignment of the feature maps. F, then, is further refined by an

Table 1 Quantitative results on **G1**

Method	Params	FPS	mIoU	mDice	*P*	*R*
UNet [9]§	7.9	11	0.471	0.597	0.672	0.617
ResUNet[11]§	8.4	15	0.572	0.690	0.745	0.725
ResUNet++ [23]§	16.2	7	0.613	0.714	0.784	0.742
FCN8 [21]§	134.3	25	0.737	0.831	0.882	0.835
HRNet [58]§	9.5	12	0.759	0.845	0.878	0.859
DoubleUNet [24]§	29.3	7.5	0.733	0.813	0.861	0.840
PSPNet [59]§	48.6	17	0.744	0.841	0.890	0.836
DeepLabv3+(ResNet50) [22]§	39.8	28	0.776	0.857	0.891	0.862
DeepLabv3+(ResNet101) [22]§	58.8	17	0.786	0.864	0.906	0.859
U-Net (ResNet34) [9]§	33.5	35	<u>0.810</u>	**0.876**	**0.943**	0.860
ColonSegNet [46]§	<u>5.0</u>	**182**	0.724	0.821	0.844	0.850
SegFormer [41]	84.7	41	0.763	0.845	0.859	<u>0.877</u>
TMPSformer (Ours)	**2.7**	<u>162</u>	**0.811**	<u>0.875</u>	<u>0.890</u>	**0.901**

Note In the experiment-1, SegFormer [41] equipped with Mit-B5 encoder results are generated using the released code at https://github.com/lucidrains/segformer-pytorch. '§' represents evaluation scores from the literature [46]. Bold figures indicate the best performance, while underlined figures mark the second-best

Table 2 Quantitative results on Kvasir and CVC-612 datasets of **G2**

	Metrics	UNet[9]‡	UNet++ [10]‡	SFA [60]‡	SegFormer[41]	UNeXt[32]	Ours
Kvasir	mDice	0.818	0.821	0.723	<u>0.836</u>	0.835	**0.872**
	mIoU	0.746	0.743	0.611	0.752	<u>0.770</u>	**0.806**
	S_α	0.858	0.862	0.782	<u>0.882</u>	0.868	**0.901**
	E_ϕ^{max}	0.893	0.910	0.849	<u>0.924</u>	0.902	**0.933**
	MAE	0.055	0.048	0.075	<u>0.042</u>	<u>0.042</u>	**0.035**
CVC-612	mDice	<u>0.823</u>	0.794	0.700	0.819	0.813	**0.896**
	mIoU	0.755	0.729	0.607	0.735	<u>0.758</u>	**0.838**
	S_α	0.889	0.873	0.793	<u>0.895</u>	0.877	**0.932**
	E_ϕ^{max}	**0.954**	0.931	0.885	0.936	0.919	**0.974**
	MAE	<u>0.019</u>	0.022	0.042	0.023	0.024	**0.010**

Note UNeXt [32] results are generated using the released code at https://github.com/jeya-maria-jose/UNeXt-pytorch. '‡' represents evaluation scores from the literature [27]. Bold figures indicate the best performance, while underlined figures mark the second-best

additional MLP layer, adeptly crafting a direct and precise prediction for the final segmentation mask, resulting in SM.

This mask is rendered with dimensions $\frac{H}{4} \times \frac{W}{4} \times N_{cls}$, where N_{cls} signifies the totality of classes under consideration for segmentation. The operations performed within these layers are precisely illustrated in Equations (16) to (17).

$$F = Linear\,(4C, C)\,(F'') \tag{16}$$

$$SM = Linear\,(C, N_{cls})\,(F) \tag{17}$$

where SM refers to the predicted mask, and Linear (C_{in}, C_{out}) refers to a linear layer where C_{in} and C_{out} represent the dimensions of the input and output vector dimensions, respectively.

4 Result

In our study, we conducted comprehensive experiments using six publicly available datasets, selected based on their proven efficacy in polyp segmentation as noted in several benchmark studies [27, 38, 46]. Our method was benchmarked against SOTA methods, employing the same datasets for a fair comparison. The comparative analysis, detailed in Tables 1, 2, 3 and 4, demonstrates our model's superior performance in terms of Dice coefficient (Dice) and Intersection over Union (IoU) metrics. Remarkably, our model achieves this with a significantly lower parameter count (2.7 M) and an unparalleled inference speed of 162 FPS at a resolution of 512 × 512 pixels. The TMPSformer's combination of high accuracy and low latency highlights its advantages over existing approaches.

Table 3 Quantitative results on **G3**

	Metrics	UNet[9]**	ACS [61]**	PNSNet [41]**	SegFormer[41]	UNeXt[32]	Ours
CVC-300-TV	maxDice	0.639	0.738	**0.840**	0.768	0.788	**0.826**
	maxIoU	0.525	0.632	**0.745**	0.650	0.699	**0.737**
	S_α	0.793	0.837	**0.909**	0.850	0.857	**0.891**
	E_ϕ	0.826	0.871	**0.921**	0.896	0.892	**0.927**
	MAE	0.027	**0.016**	**0.013**	0.028	0.018	0.018
CVC-612-V	maxDice	0.725	0.804	**0.873**	0.819	0.839	**0.903**
	maxIoU	0.610	0.712	**0.800**	0.716	0.756	**0.832**
	S_α	0.826	0.847	**0.923**	0.886	0.881	**0.937**
	E_ϕ	0.855	0.887	**0.944**	0.922	0.935	**0.966**
	MAE	0.023	0.054	**0.012**	0.019	0.018	**0.011**
CVC-612-T	maxDice	0.729	0.782	**0.860**	0.798	0.748	**0.839**
	maxIoU	0.635	0.700	**0.795**	0.701	0.661	**0.758**
	S_α	0.810	0.838	**0.903**	0.855	0.824	**0.884**
	E_ϕ	0.836	0.864	**0.903**	0.869	0.851	**0.891**
	MAE	0.058	0.053	**0.038**	0.050	0.055	**0.043**

Note '**' represents evaluation scores from the literature [31]. Bold figures indicate the best performance, while underlined figures mark the second- best

Table 4 Quantitative results on three unseen datasets (CVC-ColonDB, ETIS and CVC-300) of **G2**

	Metrics	UNet[9]‡	UNet++ [10]‡	SFA [60]‡	SegFormer[41]	UNeXt[32]	Ours
ColonDB	mDice	0.512	0.483	0.469	**0.672**	0.669	**0.690**
	mIoU	0.444	0.410	0.347	0.570	**0.588**	**0.602**
	S_α	0.712	0.691	0.634	**0.794**	0.790	**0.803**
	E_ϕ^{max}	0.776	0.760	0.765	**0.852**	**0.856**	0.850
	MAE	0.061	0.064	0.094	0.053	**0.044**	**0.046**
ETIS	mDice	0.398	0.401	0.297	**0.547**	0.485	**0.527**
	mIoU	0.335	0.344	0.217	**0.453**	0.420	**0.459**
	S_α	0.684	0.683	0.557	**0.743**	**0.739**	**0.739**
	E_ϕ^{max}	0.740	0.776	0.633	**0.807**	0.790	**0.792**
	MAE	0.036	0.035	0.109	0.051	**0.035**	**0.032**
CVC-300-T	mDice	0.710	0.707	0.467	0.750	**0.788**	**0.773**
	mIoU	0.627	0.624	0.329	0.640	**0.708**	**0.671**
	S_α	0.843	0.839	0.640	**0.858**	**0.863**	**0.858**
	E_ϕ^{max}	0.876	0.898	0.817	**0.949**	0.925	**0.936**
	MAE	0.022	0.018	0.065	**0.016**	**0.014**	0.017

Note '‡' represents evaluation scores from the literature [27]. Bold figures indicate the best performance, while underlined figures mark the second-best

4.1 Datasets

The datasets utilized in our experiments were chosen for their diversity and relevance to polyp segmentation, enabling a thorough evaluation of our model's performance and generalizability. They include Kvasir-SEG [47], CVC-612 [48], CVC-ColonDB [49], EndoScene [50], ETIS [51], and ASU-Mayo [49], segmented into three distinct groups as per existing literature [27, 38, 46], namely, group1(**G1**), group2(**G2**), and group3(**G3**).

G1 It employs 880 images from Kvasir-SEG for training, with the remaining 120 images serving as a test set to evaluate the model's proficiency on familiar datasets. Kvasir-SEG, a subset of the Kvasir dataset [52], comprises 1000 images dedicated to polyp detection.

G2 The combined dataset, consisting of Kvasir-SEG and CVC-612, is split such that 80% of the images are used for training. The remaining 20% is divided separately for each dataset: 10% of Kvasir-SEG for validation and 10% for testing, and similarly, 10% of CVC-612 for validation and 10% for testing. The CVC-612 dataset consists of 612 images sourced from 31 colonoscopy clips. To assess the model's adaptability, we tested it on three additional datasets—CVC-ColonDB, EndoScene, and ETIS—not encountered during training. CVC-ColonDB, featuring 380 images from 15 colonoscopy sequences, and ETIS, with 196 images aimed at early colorectal cancer detection, along with EndoScene (a merger of CVC-612 and CVC-300 [53]), provide a robust challenge to gauge generalizability.

G3 It utilizes the entire Kvasir-SEG dataset and positive samples from the ASU-Mayo collection for training. Images from CVC-300 and CVC-612 are divided into 60% for training, 20% for validation (CVC-300-V and CVC-612-V), and 20% for testing (CVC-300-T and CVC-612-T), allowing for an in-depth assessment of the model's learning capabilities and its effectiveness.

4.2 Implementation Details

Loss Function Configuration Our TMPSformer model, developed using the pytorch framework, is tailored for the nuanced task of polyp segmentation. To effectively train our model, we employ a hybrid loss function that combines Binary Cross Entropy (BCE) and Dice loss, leveraging their complementary strengths. BCE ensures accurate pixel-wise classification, crucial for distinguishing polyp regions from the background. However, BCE alone may not adequately address class imbalance, a common issue in medical image segmentation. Dice loss mitigates this by focusing on the overlap between the predicted segmentation and the ground truth, enhancing spatial accuracy. Together, this composite loss function optimizes overall segmentation performance by balancing pixel-wise classification accuracy with spatial overlap. The loss function, denoted as L, for a given prediction P and the target T, is mathematically formulated as Eq. (18).

$$L = 0.5 \times BCE\,(P, T) + Dice\,(P, T) \tag{18}$$

Table 5 Evaluation metrics and corresponding notions

Metrics	Representation	Evaluation
Model Size	Params	Evaluate the complexity of the segmentation model.
Inference Speed	FPS	Evaluate the clinical applicability of the segmentation model in terms of inference time during the test.
Aaverge IoU Maximum IoU [38]	mIoU maxIoU	Evaluate the similarity between two sets of data.
Aaverge Dice Maximum Dice [38]	mDice maxDice	Evaluate the overlap between two masks.
S-Measure [55]	$S\alpha$	Evaluate the similarity between prediction and GT.
Enhanced-alignment Measure [56, 57]	E_{ϕ}^{max} , E_{ϕ}	Evaluate pixel level matching and image-level statistics.
Mean Absolute Error	MAE	Evaluate the pixel-level error between the prediction and GT.
Precision	P	Evaluate the accuracy of the segmentation model in identifying the positive class among all the pixels it classified as positive.
Recall	R	Evaluate the segmentation model's ability in identifying all the positive class pixels correctly.

Hyperparamters For parameter optimization, we utilize the AdamW optimizer [54], a variant of the Adam optimizer with improved regularization. This choice is informed by its ability to combine the benefits of the Adaptive Gradient Algorithm (AdaGrad) and Root Mean Square Propagation (RMSProp), featuring a learning rate of 1×10^{-4} and momentum of 0.9. The learning rate is dynamically adjusted through a cosine annealing schedule, with the rate decaying to a minimum of 1×10^{-6} to ensure smooth convergence over training epochs. The model undergoes end-to-end training over 800 epochs, with a batch size of 16, ensuring thorough learning and generalization across the datasets. As referenced in [27, 38, 46], input dimensions are standardized to 512×512 pixels for **G1**, 352×352 pixels for **G2**, and 256×448 pixels for **G3**, catering to the unique characteristics of each dataset. The training and testing phases maintain these resolutions to preserve consistency, and both phases use the same batch size. The segmentation output is refined through a sigmoid function, producing the final prediction map that delineates polyp regions with high precision.

Metrics In our study, we adopt six metrics —model size, inference speed, average IoU, average Dice, precision, and recall— for performance evaluation in the experiments on **G1**. This follows the methodology outlined in previous research [46] to ensure a fair comparison with other methods. Similarly, in the experiments on **G2**, we adopt five metrics —average IoU, average Dice, s-measure, enhanced-alignment measure, and mean absolute error— for performance evaluation, adhering to the approach described in previous research [27] for consistency. For the experiments on **G3**, we adopt five metrics —maximum IoU, maximum Dice, s-measure, enhanced-alignment measure, and mean absolute error— for performance evaluation, based on the procedures detailed in previous research [38] to maintain fairness. All evaluation metrics are summarized in Table 5.

Environment Settings All evaluations were conducted with the following settings: (1) CPU, Intel Xeon Gold 6248. (2)

GPU, NVIDIA GeForce RTX 2080 Ti. We used Pytorch version 1.7.1 with CUDA version 11.0.

4.3 Quantitative Results

Learning Ability and Inference Speed Analysis To empirically demonstrate the robust learning capability and superior inference speed of the TMPSformer, we meticulously designed and executed three distinct experiments. The outcomes of these experiments, detailed in Tables 1, 2 and 3, underscore the effectiveness of our model in handling polyp segmentation tasks across various datasets.

Experiment 1 Illustrated in Table 1, this experiment showcases the TMPSformer's exceptional performance on **G1**. It can be observed that the light model ColonSegNet with a size being 5 M is the fastest method in terms of processing speed, achieving 182 FPS at a resolution of 512×512 pixels on a single NVIDIA RTX 2080 Ti GPU. However, the proposed TMPSformer achieves a real-time processing speed with 162 FPS and obtains the highest mean IoU of 0.811 compared with other SOTA methods, outperforming the ColonSegNet by 8.7%. Notably, the TMPSformer is the smallest among the compared models with its size only being 2.7 M, whose parameters are 46% less than the ColonSegNet (rank 2rd).

Experiment 2 As illustrated in Table 2, this phase of testing on both the Kvasir and CVC-612 datasets of **G2** further validates our model's efficacy. It can be observed that the TMPSformer performs better than the other SOTA segmentation methods in terms of mean IoU and mean Dice. The TMPSformer not only attained the highest mean IoU (0.806, 0.838) but also excelled in the mean Dice coefficient (0.872,0.896) on the two datasets mentioned above, affirming its consistency in segmentation precision across different datasets. Moreover, TMPSformer, which surpasses the second-ranked method by more than 3.6% in mean IoU on the Kvasir dataset and by over 8% on the CVC-612 dataset, also exceeds the second-ranked method by more than 3.6% in mean Dice on the Kvasir dataset and by over 7.3% on the CVC-612 dataset.

Table 6 Analysis of segmentation performances on **G1** at different resolutions

Network	Relusion	mIoU	mDice	P	R
TMPSformer	224×224	0.743	0.829	0.863	0.855
	256×256	0.744	0.828	0.861	0.858
	448×448	0.774	0.850	0.885	0.877
	512×512	**0.811**	**0.875**	**0.890**	**0.901**

Experiment 3 Detailed in Table 3, the third experiment evaluated the TMPSformer's performance on **G3**. CVC-300-TV consists of both the validation set and the test set, which include six videos in total. CVC-612-V and CVC-612-T each contain five videos. The results reveal that our model achieves remarkable performance across most metrics compared to SOTA methods, outperforming the second-ranked method, PNSNet, by a max IoU of 3.2% and a max Dice of 3% on the CVC-612-V dataset. This demonstrates its adaptability and robustness in various video image segmentation scenarios.

Assessment Across Unseen Datasets This experiment (experiment-4) focused on the model's performance across three unseen datasets within the **G2** group, which were not encountered during the initial training phase. The outcomes of this pivotal experiment are depicted in Table 4, presenting a comparative analysis against various segmentation baselines. Remarkably, TMPSformer demonstrates superior performance, excelling in both mean Dice and mean IoU metrics across all three datasets (CVC-ColonDB, ETIS, and CVC-300) compared to other SOTA methods. Notably, on the CVC-ColonDB dataset, it surpasses the second-ranked method by 1.8% in mean Dice and 1.4% in mean IoU. This not only underscores the model's robustness but also highlights its remarkable ability to adapt to new, unseen data, confirming its strong generalization capability.

4.4 Discussion

Analysis on Different Relutions To understand the effect of image resolution on the performance of our model TMPSformer, we conducted a series of experiments on **G1**, where images were resized to various resolutions during training and testing. The Kvasir-SEG dataset of **G1** includes 1,000 images with dimensions ranging from 332×487 pixels to 1920×1072 pixels. In our experiments, we resized the images to four different resolutions: 512×512, 448×448, 256×256, and 224×224. The results, as shown in Table 6, indicate that the performance metrics (mIoU, mDice) for the resolutions 512×512, 448×448, 256×256, and 224×224 are (0.811, 0.875), (0.774, 0.850), (0.744, 0.828), and (0.743, 0.829), respectively. The segmentation performance was highest at 512×512, exceeding the second-best resolution of 448×448 by 3.7% in mIoU and 2.5% in mDice. As the resolution decreases, there is a corresponding decline in segmentation performance.

The superior performance at the 512×512 resolution can be attributed to the retention of sufficient detail. Higher resolutions, such as 512×512, preserve more anatomical details and finer structures, which are crucial for accurate segmentation. However, as the resolution decreases, the images lose critical details, leading to a decline in segmentation accuracy. Additionally, lower resolutions may result in a loss of contextual information, which is essential for distinguishing between different tissue types and structures. Therefore, while lower resolutions reduce computational load, they also compromise the model's ability to accurately segment the images, as evidenced by the performance metrics. However, higher resolutions increase computational burden and reduce real-time performance, so it is essential to experiment and determine the optimal balance between efficiency and accuracy based on the specific application requirements.

Ablation Study To evaluate the specific impact of each module (FEA module and shift MLP module) within TMPSformer, we performed an ablation study on the Kvasir-SEG dataset of **G1**, detailed in Table 7.

For our ablation study, we conducted a comprehensive evaluation of the models in terms of model size (Params), real-time performance (FPS), and segmentation performance (mDice). Initially, we experimented with SegFormer equipped with the Mit-B5 encoder, which is a transformer-based model, and used it as the baseline. The baseline metrics were 84.7 M params, 41 FPS, and 0.845 mDice. Then, we tested a reduced transformer model by decreasing the number of layers in the transformer block of the baseline to reduce the number of parameters and complexity. We observed that the reduced transformer model showed a decrease in performance, with mDice dropping to 0.809 (a 3.6% reduction compared to the baseline), but it significantly reduced the number of parameters to 4.27 M and improved real-time performance to 113 FPS.

Next, based on the aforementioned reduced transformer model, we replaced the traditional self-attention module with the FEA module. Although we found a slight increase in the number of parameters to 4.38 M, the computational efficiency improved significantly, with real-time performance (FPS) increasing by 16% to 132, and segmentation accuracy slightly improving to 0.816 mDice. This improvement can be attributed to the FEA module, which leverages SRA and a memory read/write minimization technique from FlashAttention, enhancing the efficiency and performance of the segmentation process. Finally, we replaced two transformer blocks with the shift MLP blocks in our proposed model, TMPSformer. This modification resulted in a significant reduction in model parameters to 2.7 M, while both segmentation accuracy and computational efficiency improved, reaching 0.875 mDice and 162 FPS.

These ablation experiments demonstrate the effectiveness of the proposed FEA module and shift MLP module. The FEA module enhances real-time performance without compromising the segmentation accuracy of the transformer model, while the shift MLP block, through its shift-MLP operation that emphasizes local dependencies, effectively extracts crucial local detail information for segmentation performance. By combining the transformer's long-range dependencies with the shift-MLP's local dependencies, our method significantly improves precise segmentation performance.

Additionally, the baseline model, which is significantly heavier and has a maximum training batch size limited to 1 when using a single NVIDIA RTX 2080 Ti GPU, requires 72.7 h for training. In contrast, our model, TMPSformer, which has only 2.7 M parameters, allows the maximum batch size during model training to be set to 16, thereby reducing the entire training process to just 11.9 h. Due to the lightweight nature of TMPSformer, this results in a 6-fold increase in training efficiency compared to the baseline.

Qualitative Results As illustrated in Fig. 2, we provide the polyp segmentation results of our TMPSformer on the five test sets (Kvasir-SEG, CVC-612, CVC-ColonDB, ETIS, CVC-300) of **G2**.

From the quantitative results, it can be observed that CNN-based segmentation methods, such as UNet, excel at extracting local features but lack the ability to capture long-range dependencies. Consequently, they perform poorly in segmenting polyps with irregularly variable shapes and large spatial gaps, as illustrated in the subplots in the third column of the eighth row and the third column of the fifth row in Fig. 2.

Table 7 Comprehensive ablation study results on the Kvasir-SEG dataset of **G1**

Network block	Transformer block (numbers)	FEA module (numbers)	Shift MLP (numbers)	Layer (numbers)	Params	FPS	mDics
Baseline	√ (4)			(3,6,40,3)	84.7	41	0.845
Reduced Transformer	√ (4)			(3,6,2,2)	4.27	113	0.809
Reduced Transformer (FEA)	√ (4)	√ (2)		(3,6,2,2)	4.38	132	0.816
TMPSformer (Ours)	√ (2)	√ (2)	√ (2)	(3,6,2,2)	**2.7**	**162**	**0.875**

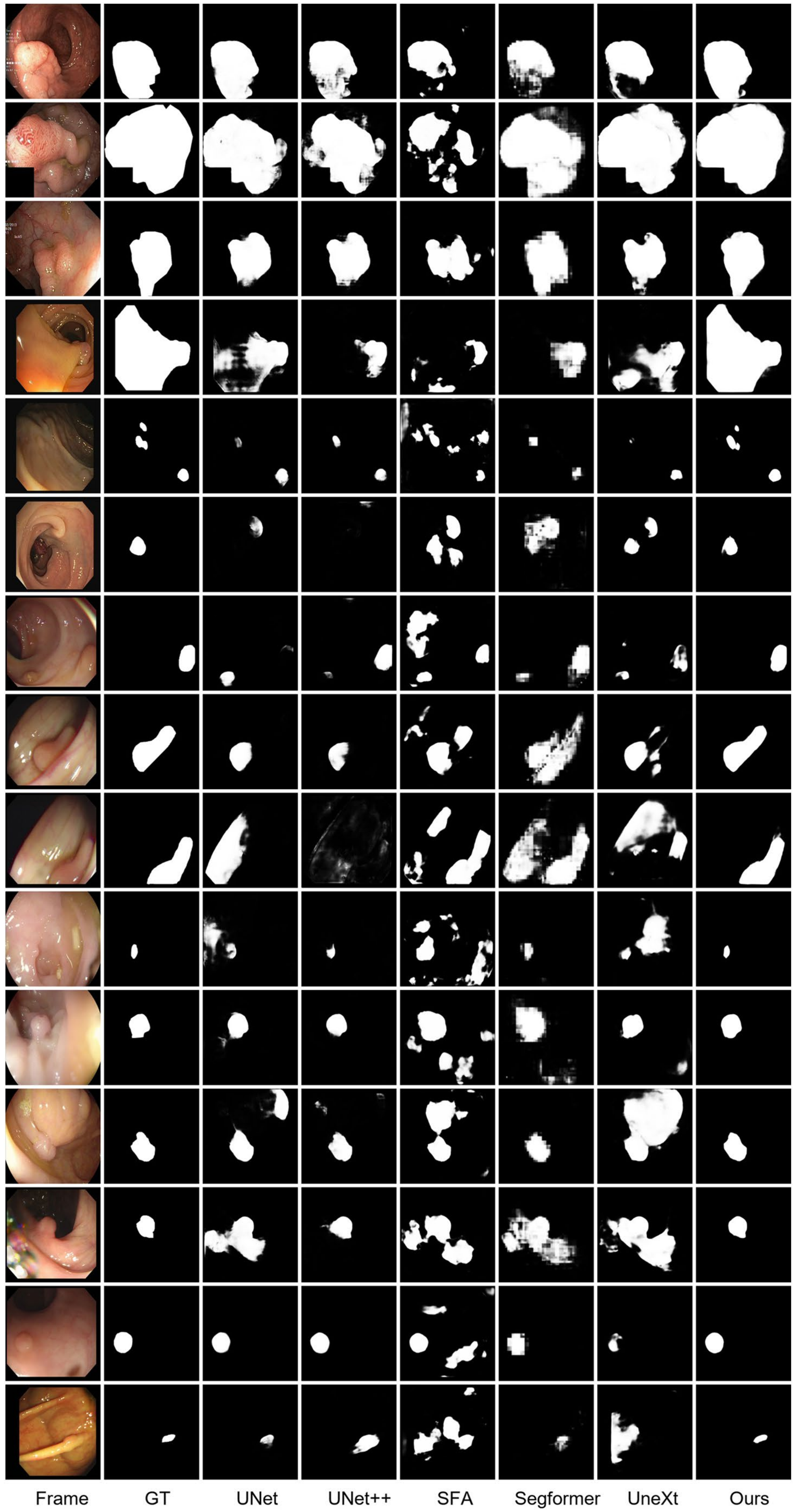

Fig. 2 Qualitative results of different methods on **G2**. Rows 1–3 correspond to Kvasir-SEG, rows 4–6 to CVC-612, rows 7–9 to CVC-ColonDB, rows 10–12 to ETIS, and rows 13–15 to CVC-300

On the other hand, Transformer-based segmentation methods, such as SegFormer, are adept at capturing long-range dependencies but are weaker in extracting local feature details. This results in poor segmentation performance for polyps with ambiguous boundaries, which refer to the boundaries between the polyps and surrounding tissues, as shown in the subplots in the sixth column of the fourth row and the sixth column of the ninth row in Fig. 2. Our model, which combines the advantages of Transformer's long-range dependency extraction and shift MLP's local feature mining, can accurately segment polyps with ambiguous boundaries and irregularly variable shapes. Our method is well-suited for real-time polyp segmentation applications, even in challenging scenarios involving varied sizes and contrasts, diverse textures, and more.

5 Conclusion

In this research, we unveiled TMPSformer, an innovative architecture designed for the meticulous segmentation of polyps from colonoscopy images. Distinguished by its exceptionally compact framework, consisting of merely 2.7 M parameters, and its remarkable inference velocity of 162 FPS at a 512×512 resolution, TMPSformer marks a substantial leap forward in the realm of efficient polyp segmentation. The model's design features a harmonious encoder that fuses Transformer and shifted MLP technologies, coupled with a streamlined all-MLP decoder. Our comprehensive evaluations have confirmed TMPSformer's prowess in outperforming the bulk of cutting-edge methods across a spectrum of demanding datasets. We foresee TMPSformer as a paradigm-shifting force in polyp segmentation, providing a robust and streamlined solution poised to greatly enhance the early detection and intervention strategies for colorectal cancer.

Author Contributions The authors confirm contribution to the paper as follows. Conceptualization: P. G. Formal analysis: P. G. Investigation: P. G. Methodology: P. G. Software: P. G. Writing-original draft: P. G. Project administration: Gp. L. Writing-review and editing: Gp. L. Supervision: Gp. L. Resources: H. L. All authors reviewed the manuscript.

Funding This work is supported by the National Natural Science Foundation of China (Grants No. 42061055).

Data Availability The datasets used and analysed during the current study available from the corresponding author on reasonable request.

Declarations

Ethics Approval This article does not contain any studies with human participants or animals performed by any of the authors.

Competing Interests The authors declare no competing interests.

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